School of Computer Science

www.cs.mcgill.ca

Course Outline

Course Name: Software Engineering Project
COMP-361 Fall 2026 D1 & Winter 2027 D2

Instructor: Joseph Vybihal

Contact Information:

Office:	ENGMC 327	Classroom D1: MCMED 1034	WF 16:00
Office Hours:	T & W 13:00	Classroom D2: WONG 1020	TTH 14:30
Discussions:	uAsk (experimental)		
Email:	Emergencies or non-course-related: joseph.vybihal@mcgill.ca		

Teaching Assistants
See the myCourses "TA & Prof Coordinates" page under the Content module.

Pre- and Co-requisites: Prerequisites: COMP 206 and COMP 250
Co-Requisites: COMP 303

Student Streams: Games Stream – Students are expected to build a game using Unity, Unreal, or C++.
All other streams – Can either work on a game or an application (see project description)

Course Description: Software development process in practice: requirement elicitation and analysis, software design, implementation, integration, test planning, and maintenance. Application of the core concepts and techniques through the realization of a large software system.

Course Objectives: This 1-year course will give students the opportunity to build a considerable software system following a rigorous software development process. Over two semesters, students will develop a system that solves a set of requirements of practical relevance. The project will take students through all the phases of a software's life cycle, from inception to final deployment and release.

During the first semester, students will attend lectures to learn the foundations of software engineering. In the second semester, lectures are replaced by progress meetings and programming.

Throughout the project, students will follow a defined process involving software engineering practices from a variety of disciplines, including requirements gathering, requirements modeling and specification, requirements analysis, software design, implementation, testing, configuration and change, and project management.

Throughout the project, students follow a specification-driven engineering approach, developing software specifications at different levels of abstraction using different modeling formalisms. The final implementation is done by the student using an object-oriented programming language of their choice.

Throughout the project, students will adhere to an Agile development process, primarily known as Scrum. A mature form of Scrum that respects structured engineering design. Each Scrum iteration will end with an artifact, model, report, or implementation.

We will use the Office365 software engineering tools provided by McGill.

Learning Outcomes: At the end of the project, students will have gained experience in several areas related to software engineering:

- Development Skills
  - Understanding a reasonably complicated problem statement
  - Analyzing the needs that an application must fulfill
  - Specifying the requirements
  - Designing a solution that fulfills the requirements
  - Implementing the solution with an object-oriented programming language
  - Code maintenance (regression, fixes, adding functionality)
  - Delivering the application on time
- Modeling Skills
  - Eliciting behavioral requirements with use case models
  - Elaborating domain models with class diagrams
  - Specifying system interfaces with object interaction diagrams
  - Specifying system protocols with use case maps
  - Specifying software design structure with class diagrams
  - Specifying software behaviour with sequence diagrams
  - Specifying the run-time environment with the deployment diagram
- Team Skills
  - Scrum management techniques
  - Contributing to the constitution and growth of a small group of developers
  - Organizing work and programming as a team
  - Communicating effectively within a group
- Tools Skills
  - Scrum tools
  - Modeling tools
  - Programming environments
    - Compilers, debuggers, profilers
  - Version control software

Texts: This course uses a **Course Pack**; see it on our myCourses page.

Other useful books: See the catalog of texts on our myCourses page.

UML Specification for download at the OMG website (http://www.omg.org/spec/UML/)

Teaching Method This course does not use lecture slides. Students are expected to read and fully understand the provided Course Pack, which can be found on our myCourses page in the Content module.

Every lecture will be a hands-on experience with the professor writing on the board or using applications on the laptop. In-class examples will be uploaded into the Handouts section of the Content module in myCourses.

Students will be given problems during class to solve in a small group or as a class activity. In-class discussions are welcome.

The course has a theory and a practical component. The theory component is evaluated through assignments and a single final exam. The practical portion is evaluated through the project. The project is divided into multiple submissions following a standard engineering workflow. In addition, mentorship is provided through Progress Meetings. The first semester is Theory and Assignments; the second semester is Project Development and Progress Meetings.

Software Development and Engineering are very disciplined environments. This course is designed similarly.

You begin forming your team and developing the project from the first month of the course. You will not do any programming during the first month, but you will plan and design. Every assignment is divided into an individual portion and a team portion. The individual portion explores some theory or technique you need to master. The team portion progresses your project. You must complete the individual portion yourself. You must complete the team portion in collaboration with your team. Everyone submits the same team portion answer, but your individual portion question is unique to you.

You cannot bring any technology or crib sheets to the final exam.

Some use of Generative AI is permitted in this course, as long as more than 50% of the submitted work is your own effort. If there are doubts, the grader will interview you, and that interview will determine your final grade.

Evaluation:

Fall Semester

- Assignment #1 (Req & Team formation)	5%
- Assignment #2 (UML1 & Spec)	10%
- Assignment #3 (UML2 & Design)	10%
- Assignment #4 (UML3 & Development)	10%
- Formal Final Exam	30%

Winter Semester

- January Progress Meeting #1 (week 2)	1%
- January Progress Demo #1 (week 4)	1%
- February Progress Meeting #2 (week 2)	2%
- February Progress Demo #2 (week 4)	2%
- March Progress Meeting #3 (week 2)	2%
- March Progress Demo #3 (week 4)	2%
- Final Demonstration of the Completed Project	10%
- Acceptance Test	10%
- Scrum Meeting Journal	5%
- Showcase	+2 points (pres)

A separate document will be handed out during class describing the project.

Late work: You will be notified in advance of assignment due dates. All assignments are due on My Courses at the indicated time and date. Late assignments will lose **10%** of their grade per day late. Assignments beyond **2 days** late will not be accepted. You may not submit assignments via e-mail without the permission of the instructor.

Additional Work: Students with grades of D, F, or J will not be given the opportunity to complete additional work to upgrade their grade.

Supplemental Exam: There will be no supplemental exam for this course.

Re-grading: Mistakes can occur when grading. Not surprisingly, requests for re-grading always involve those mistakes in which the student received fewer points than they deserved, rather than more points than they deserved. With that in mind: if you wish me to re-grade a question on an exam or assignment, I will do so. I reserve the right to re-grade other questions as well. All regrading requests must happen within 7 days after the

submission was graded.

Cheating/Collaboration: Collaboration is encouraged, but your discussions should be public in the sense that anyone, including the professor, should be allowed to listen in, and no code is shared (including through conversation). Assignments are original works created by the student alone. You are permitted and encouraged to have conversations with other students concerning the meaning of the assignments, but your work must be original. If two or more assignments are found to be identical (or portions of assignments) then all parties will lose points. This includes the student who permitted their assignment to be copied. This includes written solutions and software source code. **ChatGPT is permitted in the course as long as the assignment is 50% or more your work. This might be verified through a special interview when the professor or teaching assistant has doubts. Note: The final exam does not use any technology or crib sheets.**

Grading: All software solutions must compile with zero errors and must run to be graded. It does not need to run correctly to be graded, but it must run. If your program compiles with errors or does not run at all, then you will receive zero points. The grader will not fix your code or review the source code to give you partial credit.

Additional Information: There are no course slides. Any slides given out during class are not meant as a complete set of notes or a substitute for the Course Pack but simply constitute the focus of the lecture. Important gaps are left in the slides and filled in during class; therefore, lecture attendance is essential.

The material covered in class will supplement the readings.

Communication

My Courses**:** All official communication, including announcements, lecture material, assignments, and grades, will be found on My Courses.

Course Discussions: uAsk

Private Email: The professor and TA have private email accounts that you may also use; however, these communication channels are for personal queries. For example: if you have a problem with your grade, then email the TA who graded you directly; do not email the prof.

Appointments: Please email the one you want to communicate with directly to book an appointment outside office hours.

Office Hours: Consult the posted office hours and come to those times without an appointment.

After lecture: Some optional time will be available just after class to ask questions. I do not guarantee the length of this time since other constraints may interfere.

Right to submit in English or French written work that is to be graded

In accordance with McGill University's Charter of Students' Rights, students in this course have the right to submit in English or in French any written work that is to be graded.

Academic Integrity: *Code of Student Conduct*

McGill University values academic integrity. Therefore, all students must understand the meaning and consequences of cheating, plagiarism and other academic offenses under the Code of Student Conduct and Disciplinary

Procedures (see https://www.mcgill.ca/students/srr/academicrights/integrity for more information).

L'université McGill attache une haute importance à l'honnêteté académique. Il incombe par conséquent à tous les étudiants de comprendre ce que l'on entend par tricherie, plagiat et autres infractions académiques, ainsi que les conséquences que peuvent avoir de telles actions, selon le Code de conduite de l'étudiant et des procédures disciplinaires (pour de plus amples reseignements, veuillez consulter le site https://www.mcgill.ca/students/srr/academicrights/integrity).

Note:
In the event of extraordinary circumstances beyond the University's control, the content and/or evaluation scheme in this course is subject to change.

Tentative Course Schedule			
	LECTURE	**HOMEWORK**	
Date	**Class**	**Assignment**	**Course Pack**
UNIT 1	**COURSE INTRODUCTION**		
Sep 2	**Course introduction**. Teaching method. About the project. About the Ubisoft Games competition.		
Sep 6	**Ad hoc software development**. Basics and programming on the fly.		Chapter 1
UNIT 2	**PROFESSIONAL SOFTWARE ENGINEERING**		
	About Software Engineering & Requirements. Phases, Use Case, Storyboard, Research, Standards.	Handed out A1 Project	Chapter 2
	Project Development Strategies. Evolution and Building.		Chapter 3
	Project Management Strategies. Ownership parameters, client-in-the-loop, Effort estimation and budgeting, management style, checklists.		Chapter 4
	Planning 1. Specification document and UML 1. Design document and UML 2. Domain models and OO.	Handed out A2	Chapter 5 & 6
Oct 9 to 14	**READING WEEK**		
	Planning 2. Design document part 2 and UML 3. Development document and UML 4.		Chapter 7
	How to Engineer a Project. Separating worlds, management flow, developer flow, software tools.	Handed out A3	Chapter 8
	How to use Generative AI. Prompt engineering, structured prompts, and agentic workflows.		Chapter 9
UNIT 3	**QUALITY ASSURANCE AND PRODUCT DELIVERY**		
	Quality Assurance. Programmer unit testing with repo pipelines vs the QA Team (procedures, forms, sites, clients)	Handed out A4	Chapter 10
	Product Delivery. Deployment, Retirement, Migration, and Installation.		Chapter 11
Fall Final Exam Period	Everything covered during class, in the course pack, and in assignments.	**Formal Final Exam**	
	Winter Semester		
UNIT 4	**TEAM PROGRAMMING**		
Monthly Week 1	In-class office hours **(First class is a lecture)**		
Monthly Week 2	Progress interview with assigned grader using a checklist	Graded Progress Meeting	
Monthly Week 3	In-class office hours	**Industry Speakers**	
Monthly Week 4	Progress demo with assigned grader using a checklist	Graded Progress Demo	
Last week		**Showcase**	
Winter Final Exam Period	Appointment 1 – before exam period (final demo) Appointment 2 – during the exam period (acceptance test) (first come, first served basis) Last day oof exams – Scrum journal submission	Final Demonstration Acceptance Test Scrum Meeting Journal	

Final Exam Policy:

Regulations

Students should not make other commitments during the final exam period. Vacation plans do not constitute valid grounds for deferring or rescheduling examinations. See the Center Calendar for the regulations governing Examinations:

https://www.mcgill.ca/exams/regulations

Students must present their I.D. Card (with photo) for entrance to their examination.

Conflicts

If you are unable to write your final examination due to scheduling conflicts, you must submit a Final Exam Conflict Form with supporting documentation at least **one month** before the start of the final examination period. Late submissions will not be accepted. For details, see

https://www.mcgill.ca/exams/dates/conflicts

Exam Timetable

Examination schedules are posted at the Center and on the following page approximately 6-8 weeks before the examination period commences

https://www.mcgill.ca/exams/dates

The Center cannot provide examination dates over the telephone.

Students are responsible for all materials for the tests and exams, whether or not they are covered in class. Exams will combine all types of questions from all sources, and students may be required to integrate theoretical concepts from the text to substantiate their arguments.

No make-up tests or make-up assignments are allowed in this course.

If you are not satisfied with the grading of an assignment or mid-term test, you may request a review within 7 days of return. Indicate in writing or during a meeting with the instructor where and why you feel the marks are unjustified and give it back to your instructor for re-grading. Note that the entire assignment or mid-term test will be regraded, and your grade can go up or down (or stay the same) accordingly.

Classroom Rules: All electronic devices (cell phones and beepers) must be turned off or left on silent mode during class time.

Assignment Pickup: **All hand-ins are submitted to and picked up** from myCourses.

Email Policy: E-mail is one of the official means of communication between McGill University and its students. As with all official University communications, students are responsible for ensuring that time-critical e-mail is accessed, read, and acted upon in a timely fashion. If a student chooses to forward University e-mail to another e-mail mailbox, that student's responsible for ensuring the alternate account is viable.

Please note that, to protect students' privacy, the University will only reply to students on their McGill e-mail account.

Computing Resources: Trottier 3rd floor.

Students' Rights and Responsibilities:

Regulations and policies governing students at McGill University can be downloaded from the website:
https://www.mcgill.ca/students/srr/

Student Services and Resources:

Various services and resources, such as email access, walk safe, library access, etc., are available to students:
https://www.mcgill.ca/studentservices/

Minerva for Students: http://www.mcgill.ca/minerva-students/

Land acknowledgment:

McGill University is on land which has long served as a site of meeting and exchange amongst Indigenous peoples, including the Haudenosaunee and Anishinaabeg nations. We acknowledge and thank the diverse Indigenous people whose footsteps have marked this territory on which people of the world now gather. Please see here for more details: HYPERLINK "https://www.mcgill.ca/edu4all/other-equity-resources/traditional-territories".